Cambridge
O Level

**Cambridge International Examinations**
Cambridge Ordinary Level

**PHYSICS** **5054/31**

Paper 3 Practical Test **May/June 2017**

MARK SCHEME

Maximum Mark: 30

**Published**

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| Question | Answer | Marks |
|---|---|---|
| 1(a) | evidence of repeats used to obtain $t$ seen in (a) or (b) | **B1** |
| | $t_1 = 4 \pm 1$ s or centre value $\pm 1$ s | **B1** |
| 1(b) | any $t_1 < t_2$ and correct unit seen in (a) or (b) | **B1** |
| 1(c) | $t_1 / t_2$ in range 1.80 to 2.20 | **B1** |
| | ratio correct , given to 2 or 3 s.f. with no unit | **B1** |

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| Question | Answer | Marks |
|---|---|---|
| 2(a) | length given to nearest mm, with unit, and to centre value ± 1.0 cm<br>if no centre value (from supervisor or candidate scripts), allow in range 8.0–11.0 cm | **B1** |
| 2(b)(i) | vary / measure the distance between the lamp and the object / along XZ(owtte) (1) | **B2** |
| | compare / measure (the length of) the shadow (along PZQ) (owtte) (1) | |
| | allow 1 mark for 'move object and measure shadow'<br>max 1 mark if lamp moved | |
| 2(b)(ii) | *y*-axis labelled: length of shadow / PZQ, units not required | **C1** |
| | straight line decreasing or curve decreasing | **A1** |

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| Question | Answer | Marks |
|---|---|---|
| 3(a)(i) | centre value ± 0.5 V or 1.5 to 3.5 V, readings to at least 0.1 V, with unit | **B1** |
| 3(a)(ii) | centre value ± 0.5 V or 1.5 to 2.2 V, readings to at least 0.1 V, with unit | **B1** |
| 3(a)(iii) | $V_{AC}$ in range ($V_{AB}$ + $V_{BC}$) ± 0.2<br>allow centre value ± 0.2 | **B1** |
| 3(b) | current, I, with unit e.g. 0.0089 A ( i.e. (a)(i) / 330) | **B1** |
| 3(c) | resistance $R_L$ to 2 or 3 s.f. with unit e.g. 224 Ω<br>allow centre value ± 30 ohms or ecf (a)(ii) / (b) in range 150–250 ohms if no centre value available | **B1** |

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| Question | Answer | Marks |
|---|---|---|
| 4(a) | room temperature close to centre value if given or in range 15–45 °C<br>unit required, accept correct symbol or 'Celsius' | **B1** |
| 4(b)(i) | current $I$ less than 1 A to at least 1 d.p.<br>unit required | **B1** |
| 4(b)(ii) | sensible value for $P$ (b)(i)$^2 \times 4$ in range 0.1 W up to 4 W | **B1** |
| 4(c)(i) | headings for table, with units: temperature / °C time / s | **B1** |
| 4(c)(iii) | correct trend in results: temperature increases with time | **C1** |
| | temperature rises increase by 2.0 °C from $\theta_r$ | **C1** |
| | all times and temperatures present, starting with $\theta = (\theta_r + 2)$ (additions all performed) and finishing at $\theta = (\theta_r + 18)$ | **A1** |
| 4(d) | graph: axes labelled, with units, temperature on the $x$-axis | **B1** |
| | suitable scale, not based on 3, 6, 7 etc. with plotted data occupying at least half the page in both directions; the origin may be included | **B1** |
| | all the data in table plotted, points plotted correctly on a scale that is easy to follow at least two points checked, points must be within ½ small square of the correct position | **B1** |
| | best fit, fine line (curved or straight) and fine points or crosses | **B1** |

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| Question | Answer | Marks |
|---|---|---|
| 4(e) | tangent drawn to the curved best line at $(\theta_r + 11)$ °C | **M1** |
| | correct calculation of the gradient, given to 2 / 3 s.f., ignore any units | **A1** |
| | a large gradient triangle used, based on the tangent used | **B1** |
| | **or** | |
| | (in cases where there is no obvious curve indicated by the plotted data and a best fit straight line has been drawn then allow calculation of the gradient using a gradient triangle including $(\theta_r + 11)$ °C) | **(M1)** |
| | correct calculation of the gradient, given to 2 / 3 s.f., ignore any units | **(A1)** |
| | a large gradient triangle used centred near or on $(\theta_r + 11)$ °C | **(B1)** |

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